

From Local Curvature to the Weil Functional:

An Explicit Construction

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Research Note · August 2026 · v37

Abstract

Building on the curvature decomposition established in [1], we construct an explicit family of admissible test functions $g_{\sigma,\varepsilon}^* \in S_{\text{ad}}$ for the Weil explicit formula and identify a formal diagonal localisation pattern that targets the local curvature $H_{\text{local}}(\sigma, \kappa)$ on the prime side. A renormalised family of prime weights $c_p^{\text{ren}} = f_p^{1/2} > 0$ is introduced, and the associated diagonal energy $D = \sum_p (c_p^{\text{ren}})^2 \approx 9.470$ is shown to converge. The renormalised local curvature is shown to be of lower order than $(\log \kappa)^2$ and to exhibit a sawtooth structure related to classical Mertens-type prime-sum asymptotics. No RH, GUE, or Hilbert–Pólya input is used. No theorem or result in this paper is conditional on positive answers to these open problems: the paper isolates three open questions for the Weil-normalised embedding.

What is proved. The antisymmetrised Gaussian family $g_{\sigma,\varepsilon}^*$ belongs to the admissible class S_{ad} used in Lagarias [3] (Lemma 2.1). The renormalised prime weight $c_p^{\text{ren}} := (V_p(\frac{1}{2}) - \frac{4(\log p)^2}{p})^{1/2}$ is strictly positive (Lemma 4.1), and the diagonal energy series $D = \sum_p (c_p^{\text{ren}})^2$ converges absolutely (Lemma 4.2). The normalised Gaussian correlation K_ε localises the diagonal first-prime contribution exactly to $V_p(\sigma)$ by an algebraic identity (Observation 3.2).

What is numerical. The diagonal energy computation gives $\sum_{p < 10^7} f_p = 9.4702$, with tail $\sum_{p \geq 10^7} f_p < 2.7 \times 10^{-4}$, hence $D \approx 9.470$ to three decimal places. Separately, at zero-sum reference parameters $(\varepsilon, N) = (0.05, 100)$, direct grid evaluation of the prime-only renormalised quantities at $\kappa \in \{10, 20, 29, 50, 100, 200, 500, 1000\}$ shows that $Z_{\text{ren}} \geq H_{\text{local}}^{\text{pr}}$ holds at every tested point except $\kappa = 20$, where $Z_{\text{ren}} - H_{\text{local}}^{\text{pr}} = -0.988$, with the ratio $Z_{\text{ren}}/H_{\text{local}}^{\text{pr}}$ equal to about 1.5 at $\kappa = 29$ and about 10.1 at $\kappa = 1000$ (prime-only comparison; the archimedean term $\psi_1(\frac{1}{2}) \approx 2.150$ enters separately). No claim is made about κ outside the eight tested points.

What remains open. Three questions separate the present construction from a full prime-side derivation of Weil positivity for this family (§ 7). Problem 7.1 (*Weil-normalised embedding*) asks whether the diagonal localisation of Observation 3.2 can be implemented inside the Lagarias–Weil functional, including the correct Gaussian rescaling, the archimedean contribution, and the zero-side contribution in its unconditional form. Problem 7.2 (*Spectral inequality*) asks whether, for every fixed cutoff κ , there exists $\varepsilon_0(\kappa) > 0$ such that $\mathcal{Z}_W(g_{1/2,\varepsilon}^*) \geq H_{\text{local}}(\frac{1}{2}, \kappa)$ holds unconditionally for all $0 < \varepsilon < \varepsilon_0(\kappa)$. Problem 7.3 (*Off-diagonal control*) asks for an unconditional bound on the cross-terms for the proxy family.

Structure. The formal core in §§ 2–5 is non-circular: the analytic statements are labelled as proved, the finite computations are labelled as numerical, and the Weil-normalised embedding questions are isolated as open problems in § 7. No use of the Riemann Hypothesis, the GUE conjecture, or the Hilbert–Pólya postulate is made anywhere in §§ 2–5.

Notation and Conventions

Critical line. Throughout, the critical line means $\Re(s) = \frac{1}{2}$. The construction parameter $\sigma > 0$ is evaluated at $\sigma = \frac{1}{2}$ in the renormalised analysis unless stated otherwise.

Global parameters. The reference values are $\kappa \geq 2$ (prime cutoff, varied across the comparison grid), $\varepsilon = 0.05$ (Gaussian smoothing parameter), and $N = 100$ (number of Riemann ordinates γ_k used in the zero-sum evaluations of § 6). Values are fixed within each statement and explicitly varied otherwise.

Phasor convention. Throughout this paper the prime phasor is $\sin(\gamma_k \log p)$ evaluated at $\sigma = \frac{1}{2}$; no additional σ -factor enters the Fourier side. The Fourier transform $\widehat{g}$ of a test function g uses the convention of Lagarias [3]; the admissible test function $g_{\sigma,\varepsilon}^*$ is defined in log-prime position space in § 2; its Fourier transform is derived there.

Main objects. The local curvature, imported from [1], is

$$H_{\text{local}}(\sigma, \kappa) = \psi_1(\sigma) + \sum_{p \leq \kappa} V_p(\sigma), \quad V_p(\sigma) = 4(\log p)^2 \frac{p^{-2\sigma}}{(1 - p^{-2\sigma})^2}.$$

The Gaussian bump is $\varphi_\varepsilon(x) = (\varepsilon\sqrt{2\pi})^{-1}e^{-x^2/(2\varepsilon^2)}$. The raw weight at level σ is $c_{p,\sigma} = V_p(\sigma)^{1/2} \cdot p^{1/4}/(\log p)^{1/2}$, and the antisymmetrised admissible test function $g_{\sigma,\varepsilon}^* \in S_{\text{ad}}$ is constructed in § 2. The Weil quadratic functional is denoted W ; its zero-side contribution in the Lagarias normalisation [3] is denoted $\mathcal{Z}_W(g)$. The *numerical ordinate proxy* is the finite positive sum $Z_{\text{ord}}(g; N) := \sum_{k=1}^N |\widehat{g}(\gamma_k)|^2$ over the first N verified critical-line ordinates γ_k . The unconditional Lagarias–Weil zero-side contribution $\mathcal{Z}_W(g)$ is distinct from Z_{ord} ; the two objects are not identified without an explicit embedding statement (Problem 7.1). At $\sigma = \frac{1}{2}$ the renormalised prime weight is $c_p^{\text{ren}} = f_p^{1/2}$, where $f_p = V_p(\frac{1}{2}) - 4(\log p)^2/p$, and the diagonal energy is $D = \sum_p (c_p^{\text{ren}})^2$. The renormalised local curvature is $H_{\text{local}}^{\text{ren}}(\frac{1}{2}, \kappa) = H_{\text{local}}(\frac{1}{2}, \kappa) - 2(\log \kappa)^2$.

Analytical foundations. The construction uses the Weil explicit formula [4] in the Lagarias admissible formulation [3] and the prime number theorem for the asymptotic scale of H_{local} .

Numerical computations. All numerical results were computed with `mpmath` at 50 decimal digits of working precision; Riemann zeros γ_k were evaluated via `mpmath.zetazero`. Verification code is available at <https://github.com/utehrani/analysislab-nt>.

The diagonal energy $D = \sum_p f_p \approx 9.470$ defined here is specific to the renormalised prime-side construction of this paper. It is the convergent prime-power-tail constant in (6) and should not be read as a general trace-formula diagonal term. The renormalised weight $c_p^{\text{ren}} = f_p^{1/2}$ used in the Weil functional differs in normalisation from the unrenormalised weight $c_{p,\sigma}$ used in the construction of $g_{\sigma,\varepsilon}^*$.

1 Introduction

1.1 Background and Motivation

This paper is the second in a series developing an operator-theoretic framework for studying the distribution of the non-trivial zeros of the Riemann zeta function $\zeta(s)$. The central organising principle of the series is that prime numbers mediate coupling between zero ordinates, and that explicit finite-cutoff constructions on the prime side carry arithmetic information accessible by classical means.

Paper 1 [1] introduced the curvature field

$$H_\xi(\sigma, t) := \frac{d^2}{d\sigma^2} \log |\xi(\sigma + it)|^2 = H_{\text{local}}(\sigma, \kappa) + H_{\text{dual}}(\sigma, t, \kappa),$$

where $H_{\text{dual}} = H_\xi - H_{\text{local}}$ is the bookkeeping remainder defined as the finite-cutoff difference (cf. [1]), and where the truncated local curvature

$$H_{\text{local}}(\sigma, \kappa) = \psi_1(\sigma) + \sum_{p \leq \kappa} V_p(\sigma)$$

contains the archimedean term and the finite-place contributions. The prime-side pieces satisfy $V_p(\sigma) \geq 0$, and on the critical line one has $H_{\text{local}}(\frac{1}{2}, \kappa) \asymp (\log \kappa)^2$.

The present paper addresses a concrete part of the central question raised in the Open Problems section of [1, § 8]: whether the local positivity and critical divergence of H_{local} can be organised into an explicit admissible test-function construction whose prime-side term in the Weil functional matches the local curvature. The present paper addresses the first, purely algebraic part of this question. We construct an explicit family $g_{\sigma, \varepsilon}^* \in \mathcal{S}_{\text{ad}}$ and show that its normalised Gaussian diagonal contribution reproduces each local curvature term $V_p(\sigma)$ at fixed cutoff (Observation 3.2). The implementation of this localisation inside the fully normalised Lagarias–Weil functional — including Gaussian rescaling, the archimedean term, and the unconditional zero-side contribution — is left as Problem 7.1.

The construction $g_{\sigma, \varepsilon}^* \in \mathcal{S}_{\text{ad}}$ developed here should be distinguished from the established operator-theoretic programmes addressing the Riemann Hypothesis. It is not a Hilbert–Pólya spectral realisation in the sense of Connes [8] or Berry–Keating [9], nor a de Branges structure-function construction [10]; it is an explicit antisymmetrised Gaussian smearing of prime delta masses with arithmetic weights, closest in spirit to Hilbert-space reformulations of the explicit formula such as those of Burnol [11]. In the formal curvature dictionary of [1], the local curvature $H_{\text{local}}(\sigma, \kappa)$ is the finite-cutoff local object that the present construction is designed to realise on the prime side of a Weil-functional framework. This paper proves the normalised diagonal localisation of the finite-prime contribution (Observation 3.2); the fully normalised Lagarias–Weil embedding, including the archimedean and zero-side terms, is left as Problem 7.1. The curvature label reflects the second-derivative origin of $V_p(\sigma)$ in $\log |\xi|^2$ identified in [1] and is not used in any further geometric sense. The terms “prime–zero coupling” and “curvature field” used informally in the series are translations of the explicit-formula duality and of $\partial_\sigma^2 \log |\xi|^2$

respectively, not new mathematical structures.

The construction is set within the Lagarias formulation of Weil positivity [3], which characterises the Riemann Hypothesis as positivity of the Weil functional $W(f * \check{f})$ on a suitable class of test functions, building on the Weil explicit formula [4] and the Li positivity criterion [5]. The renormalised diagonal scale $D = \sum_p (c_p^{\text{ren}})^2$ encountered in §4 is closely related, via the prime number theorem and Mertens' classical estimates [6], to the average behaviour of the local curvature; this connection is made precise in §5.

1.2 Main Results

The paper contains three main components.

The first is the explicit construction of the test family $g_{\sigma,\varepsilon}^*$ and the proof that it lies in the admissible class S_{ad} used by Lagarias [3]. The construction proceeds in two steps: a Gaussian smearing of delta masses at $\log p$ with weights $c_{p,\sigma}$, followed by antisymmetrisation enforcing the vanishing condition at the origin (§2). The admissibility lemma (Lemma 2.1) is a direct verification.

The second is the formal diagonal localisation (Observation 3.2): the normalised Gaussian correlation K_ε targets the local curvature $V_p(\sigma)$ at each prime by an algebraic identity in the weights $c_{p,\sigma}$, while all off-diagonal and higher prime-power contributions are exponentially suppressed as $\varepsilon \rightarrow 0$ at fixed κ . The precise embedding of this localisation into the Lagarias–Weil functional — including the correct Gaussian rescaling, the zero-side contribution, and the archimedean term — is formulated as an open problem (Problem 7.1).

The third is the renormalised analysis at the critical line (§§4–5). At $\sigma = \frac{1}{2}$ the auxiliary quantity $f_p = V_p(\frac{1}{2}) - 4(\log p)^2/p$ is shown to satisfy the closed form

$$f_p = \frac{4(\log p)^2(2p-1)}{p(p-1)^2} > 0 \quad (p \geq 2)$$

(Lemma 4.1), so that $c_p^{\text{ren}} = f_p^{1/2}$ is real and strictly positive. The diagonal energy $D = \sum_p (c_p^{\text{ren}})^2$ converges absolutely by comparison with $\sum_n (\log n)^2/n^2$ (Lemma 4.2) and evaluates to $D \approx 9.470$. The renormalised local curvature $H_{\text{local}}^{\text{ren}}(\frac{1}{2}, \kappa) = H_{\text{local}}(\frac{1}{2}, \kappa) - 2(\log \kappa)^2$ is of lower order than $(\log \kappa)^2$ and exhibits a sawtooth structure: a positive jump $V_p(\frac{1}{2})$ at each new prime $p \leq \kappa$, decreasing continuously between primes as $2(\log \kappa)^2$ grows. By contrast, the prime-power-tail energy $H_{\text{tail}}(\kappa) := \sum_{p \leq \kappa} f_p$ jumps by f_p at each new prime and converges monotonically to D . The classical estimate $\sum_{p \leq \kappa} (\log p)/p = \log \kappa + O(1)$ controls the average drift, with the Mertens constant γ_M appearing in the related harmonic prime-sum estimate $\sum_{p \leq \kappa} 1/p = \log \log \kappa + \gamma_M + o(1)$.

The numerical comparison of §6 finds that $Z_{\text{ren}} \geq H_{\text{local}}^{\text{pr}}$ holds at every point of the eight-point test grid except $\kappa = 20$, with the ratio equal to about 1.5 at $\kappa = 29$ and about 10.1 at $\kappa = 1000$. Three open problems remain: the Weil-normalised embedding (Problem 7.1), the unconditional spectral inequality $\mathcal{Z}_W(g_{1/2,\varepsilon}^*) \geq H_{\text{local}}(\frac{1}{2}, \kappa)$ (Problem 7.2), and the off-diagonal control of the

cross-terms (Problem 7.3).

1.3 Reader Contract

This paper *proves*, by direct construction and classical identities, the admissibility of the test family (Lemma 2.1), the strict positivity of the renormalised weight $c_p^{\text{ren}} = f_p^{1/2} > 0$ (Lemma 4.1), and the absolute convergence of the diagonal energy series $D = \sum_p (c_p^{\text{ren}})^2$ (Lemma 4.2). The normalised Gaussian correlation K_ε localises the diagonal first-prime contribution to $V_p(\sigma)$ by an algebraic identity (Observation 3.2). Lemma 5.1 and Observation 5.2 establish the renormalised scale and sawtooth structure of $H_{\text{local}}^{\text{ren}}$. These are unconditional mathematical statements.

This paper *establishes numerically*, at reference parameters, the diagonal energy value $D \approx 9.470$ (Observation 4.3) and the finite-grid comparison $Z_{\text{ren}} \geq H_{\text{local}}^{\text{pr}}$ on the eight-point grid $\kappa \in \{10, 20, 29, 50, 100, 200, 500, 1000\}$ at every tested point except $\kappa = 20$ (Observation 6.1). Both numerical statements are finite computations from the stated parameters and are labelled [numerical] in their theorem headings.

This paper *leaves open* three well-posed questions named in § 7: Problem 7.1 (Weil-normalised embedding), Problem 7.2 (Spectral inequality), and Problem 7.3 (Off-diagonal control). All three are formulated entirely within classical analytic number theory.

2 Construction of the admissible test family

Recall from [1, § 2, finite-place contribution] that

$$V_p(\sigma) = 4(\log p)^2 \frac{p^{-2\sigma}}{(1 - p^{-2\sigma})^2}.$$

Define the raw weight

$$c_{p,\sigma} := V_p(\sigma)^{1/2} \cdot \frac{p^{1/4}}{(\log p)^{1/2}},$$

and let

$$\varphi_\varepsilon(x) = \frac{1}{\varepsilon\sqrt{2\pi}} e^{-x^2/(2\varepsilon^2)}$$

be the standard Gaussian bump. For fixed κ set

$$f_{\sigma,\varepsilon}(x) := \sum_{p \leq \kappa} c_{p,\sigma} \varphi_\varepsilon(x - \log p).$$

The admissible test function is obtained by antisymmetrisation:

$$g_{\sigma,\varepsilon}^*(x) := f_{\sigma,\varepsilon}(x) - f_{\sigma,\varepsilon}(-x). \tag{1}$$

Under the Fourier convention $\widehat{h}(t) = \int_{\mathbb{R}} h(x) e^{-itx} dx$, the Gaussian Fourier pair $\widehat{\varphi}_\varepsilon(t) = e^{-\varepsilon^2 t^2/2}$ gives

$$\widehat{g}_{\sigma,\varepsilon}^*(t) = -2i e^{-\varepsilon^2 t^2/2} \sum_{p \leq \kappa} c_{p,\sigma} \sin(t \log p), \quad (2)$$

and hence

$$|\widehat{g}_{\sigma,\varepsilon}^*(\gamma_k)|^2 = 4 e^{-\varepsilon^2 \gamma_k^2} \left(\sum_{p \leq \kappa} c_{p,\sigma} \sin(\gamma_k \log p) \right)^2. \quad (3)$$

The admissibility class S_{ad} used here is the one appearing in Lagarias' formulation of Weil positivity [3], built on Weil's explicit formula [4]. The foundational explicit-formula and test-function framework is closely related to Barner's formulation of Weil's explicit formula [14]. Avdispahić–Smaajlović extend the formula to broader test-function classes [18]; our construction remains inside the Lagarias admissible class. A test function g is admissible if g and $\widehat{g}$ are both Schwartz, $g(0) = \widehat{g}(0) = 0$, and the prime-side and zero-side expressions in the Weil explicit formula converge absolutely. Explicit admissible families in the Burnol–Sonin framework are studied in [11, 16]. To the best of our knowledge, the Gaussian family $g_{\sigma,\varepsilon}^*$ used here gives a prime-indexed realisation with closed-form weights specifically targeting H_{local} . For fixed κ , we write $g_{\sigma,\varepsilon}^*$ for $g_{\sigma,\varepsilon,\kappa}^*$, suppressing κ when no confusion arises.

Lemma 2.1 (Admissibility; proved). *For every finite cutoff $\kappa \geq 2$, every $\sigma > 0$, and every $\varepsilon > 0$, the function $g_{\sigma,\varepsilon,\kappa}^*$ belongs to the admissible class S_{ad} used by Lagarias [3] in the formulation of Weil positivity.*

Proof. The conditions of Lagarias [3] — Schwartz class (g^* and $\widehat{g}^*$ rapidly decreasing), vanishing at the origin ($\widehat{g}^*(0) = g^*(0) = 0$), and absolute convergence of the prime-side and zero-side sums — are all satisfied. The Gaussian bumps φ_ε are Schwartz, and a finite linear combination of translated Schwartz functions is again Schwartz; antisymmetrisation preserves the Schwartz class. Hence $g_{\sigma,\varepsilon}^*$ and $\widehat{g}_{\sigma,\varepsilon}^*$ are rapidly decreasing. By antisymmetry, $\widehat{g}_{\sigma,\varepsilon}^*(0) = 0$ and $g_{\sigma,\varepsilon}^*(0) = 0$. The Gaussian decay $e^{-\varepsilon^2 \gamma_k^2}$ in the Fourier representation ensures absolute convergence of $\sum_k |\widehat{g}_{\sigma,\varepsilon}^*(\gamma_k)|^2$ over all zero ordinates γ_k . More precisely, for a non-trivial zero $\rho = \beta + i\gamma$ with $0 < \beta < 1$, let $z_\rho = (\rho - \frac{1}{2})/i = \gamma - i(\beta - \frac{1}{2})$. Since $|\text{Im } z_\rho| \leq \frac{1}{2}$, the Fourier formula (2) and the finiteness of the prime sum give a constant $C_{\sigma,\varepsilon,\kappa}$ such that $|\widehat{g}_{\sigma,\varepsilon}^*(z_\rho)| \leq C_{\sigma,\varepsilon,\kappa} e^{-\varepsilon^2 \gamma^2/2}$. The Riemann–von Mangoldt bound $N(T) = O(T \log T)$ then implies absolute convergence of the zero-side sum. No critical-line localisation is used. For the prime side, use the explicit Gaussian factor in (2), not only the Schwartz property. For $u = m \log p$,

$$|\widehat{g}_{\sigma,\varepsilon}^*(u)| \leq C_{\sigma,\kappa} e^{-\varepsilon^2 u^2/2} = C_{\sigma,\kappa} e^{-\varepsilon^2 m^2 (\log p)^2/2}.$$

Hence for every $B > 0$ there is C_B such that $|\widehat{g}_{\sigma,\varepsilon}^*(m \log p)| \leq C_B p^{-Bm}$. Choosing B large enough for the Lagarias prime-power weights gives absolute convergence of the double prime-power sum. $\square$

3 Formal prime-side localisation and the open Weil bridge

Define the *normalised Gaussian correlation*

$$K_\varepsilon(x) := \frac{[\varphi_\varepsilon * \check{\varphi}_\varepsilon](x)}{[\varphi_\varepsilon * \check{\varphi}_\varepsilon](0)} = \exp\left(-\frac{x^2}{4\varepsilon^2}\right).$$

Thus $K_\varepsilon(0) = 1$ and $K_\varepsilon(x) \rightarrow 0$ exponentially for every fixed $x \neq 0$ as $\varepsilon \rightarrow 0$ (Lemma 3.1).

Lemma 3.1 (Gaussian localisation; proved). *For the family $g_{\sigma,\varepsilon}^*$ of § 2, at fixed (σ, κ) as $\varepsilon \rightarrow 0$, $K_\varepsilon(x) \rightarrow 0$ exponentially for every fixed $x \neq 0$.*

Proof. By definition, $K_\varepsilon(x) = e^{-x^2/(4\varepsilon^2)}$, so for fixed $x \neq 0$ and $\varepsilon \rightarrow 0$ the exponent $-x^2/(4\varepsilon^2) \rightarrow -\infty$. $\square$

Observation 3.2 (Formal diagonal localisation; proved). *With the normalised Gaussian correlation K_ε and the weights $c_{p,\sigma} = V_p(\sigma)^{1/2} p^{1/4}/(\log p)^{1/2}$, the diagonal first-prime contribution satisfies the algebraic identity*

$$\frac{\log p}{\sqrt{p}} c_{p,\sigma}^2 K_\varepsilon(0) = V_p(\sigma).$$

Every fixed off-diagonal prime contribution ($p \neq q$) and every fixed higher-prime-power contribution with non-zero logarithmic separation satisfies $K_\varepsilon(\cdot) \rightarrow 0$ exponentially as $\varepsilon \rightarrow 0$ at fixed κ (Lemma 3.1). No uniform summation over the full prime-power side is asserted here; that requires the separate tail/dominated-convergence argument isolated in Problem 7.1.

Proof. By the definition of $c_{p,\sigma}$,

$$\frac{\log p}{\sqrt{p}} c_{p,\sigma}^2 = \frac{\log p}{\sqrt{p}} \cdot V_p(\sigma) \frac{\sqrt{p}}{\log p} = V_p(\sigma).$$

Since $K_\varepsilon(0) = 1$, the identity follows. The off-diagonal and prime-power decay is immediate from $K_\varepsilon(x) = e^{-x^2/(4\varepsilon^2)} \rightarrow 0$ for $x \neq 0$. $\square$

Observation 3.2 shows that the construction is structured to target $H_{\text{local}}(\sigma, \kappa)$ on the prime side: the weights $c_{p,\sigma}$ are precisely designed so that the normalised diagonal contribution reproduces $V_p(\sigma)$ at $K_\varepsilon(0) = 1$, while all contributions with non-zero logarithmic separation are exponentially suppressed. The identity holds by construction: the content of Observation 3.2 is the existence of a Schwartz-class admissible family in S_{ad} with this localisation property. The normalisation factor $p^{1/4}$ in the weights $c_{p,\sigma}$ is consistent with the Haran–Connes local p -adic normalisation [17, 8]. The precise embedding of this localisation into the Lagarias–Weil functional [3] — including the correct Gaussian rescaling, the zero-side contribution, and the archimedean term — is an open problem (Problem 7.1).

More explicitly, at fixed finite κ , set

$$\delta_\kappa := \min_{\substack{p \neq q \\ p, q \leq \kappa}} |\log p - \log q| > 0.$$

Then every fixed off-diagonal contribution is $O(\exp(-\delta_\kappa^2/(4\varepsilon^2)))$ as $\varepsilon \rightarrow 0$. For the full prime-power side of the Weil functional, where infinitely many prime powers occur, this termwise localisation must be combined with a separate tail/dominated-convergence argument; that global embedding is part of Problem 7.1.

Remark. The vanishing condition $\widehat{g}_{\sigma,\varepsilon}^*(0) = 0$ (Lemma 2.1) ensures that the constant term in the Lagarias decomposition of the Weil quadratic functional vanishes, leaving only the prime-side, archimedean, and zero-side contributions.

4 Renormalised weights and convergence

At $\sigma = \frac{1}{2}$ define

$$f_p := V_p(\tfrac{1}{2}) - \frac{4(\log p)^2}{p}.$$

A direct calculation gives

$$f_p = \frac{4(\log p)^2(2p-1)}{p(p-1)^2} > 0 \quad (p \geq 2). \quad (4)$$

Lemma 4.1 (Positivity of the renormalised weight; proved). *For every prime $p \geq 2$, the quantity f_p in (4) is strictly positive.*

Proof. The numerator $2p-1 > 0$ and the denominator $p(p-1)^2 > 0$ for every prime $p \geq 2$. $\square$

Remark (Prime-power-tail interpretation). *Using the Lambert-series identity $V_p(\frac{1}{2}) = 4(\log p)^2 \sum_{m \geq 1} m p^{-m}$, one has*

$$f_p = 4(\log p)^2 \sum_{m \geq 2} m p^{-m}.$$

Thus f_p isolates the prime-power tail of $V_p(\frac{1}{2})$ after removing the leading $m = 1$ term $4(\log p)^2/p$.

The renormalised weight is then simply

$$c_p^{\text{ren}} := f_p^{1/2} > 0, \quad (5)$$

so that $(c_p^{\text{ren}})^2 = f_p$ exactly. Numerical values for small primes:

p	$V_p(\frac{1}{2})$	$4(\log p)^2/p$	$f_p = (c_p^{\text{ren}})^2$
2	3.844	0.961	2.883
3	3.621	1.609	2.012
5	3.238	2.072	1.166
7	2.945	2.164	0.781
11	2.530	2.091	0.439

Lemma 4.2 (Convergence of the diagonal energy; proved). *The series*

$$D := \sum_p (c_p^{\text{ren}})^2 = \sum_p \frac{4(\log p)^2(2p-1)}{p(p-1)^2} \quad (6)$$

converges absolutely.

Proof. For large p , $(c_p^{\text{ren}})^2 \sim 8(\log p)^2/p^2$. Since $\sum_n (\log n)^2/n^2 < \infty$, absolute convergence follows by comparison with this dominating series. $\square$

Observation 4.3 (Diagonal energy value; numerical). *The partial sum over primes below 10^7 gives*

$$\sum_{p < 10^7} f_p = 9.4702318356.$$

For $p \geq 10^7$, the closed form (4) satisfies $f_p < 9(\log p)^2/p^2$, so the tail is bounded by

$$\sum_{p \geq 10^7} f_p < \sum_{n \geq 10^7} \frac{9(\log n)^2}{n^2} < 2.7 \times 10^{-4}.$$

Therefore $D = \sum_p f_p \in (9.4702318, 9.4704965)$. In particular $D \approx 9.470$ to three decimal places (see also [2]).

Remark (Prime-zeta-tail identity). *Summing the prime-power-tail representation of f_p (Remark 4) over all primes gives*

$$D = \sum_p f_p = 4 \sum_{m \geq 2} m \sum_p (\log p)^2 p^{-m}.$$

The inner prime sums $\sum_p (\log p)^2 p^{-m}$ converge absolutely for $m \geq 2$ and equal $P''(m)$, the second derivative of the prime-zeta function $P(s) = \sum_p p^{-s}$, since $P''(s) = \sum_p (\log p)^2 p^{-s}$. This identifies D as the global prime-zeta-tail constant of the renormalisation.

5 The renormalised scale and Mertens-type prime sums

Lemma 5.1 (Renormalised local scale; proved). *As $\kappa \rightarrow \infty$,*

$$H_{\text{local}}^{\text{ren}}(\tfrac{1}{2}, \kappa) := H_{\text{local}}(\tfrac{1}{2}, \kappa) - 2(\log \kappa)^2 = O(\log \kappa).$$

In particular, $H_{\text{local}}^{\text{ren}}(\tfrac{1}{2}, \kappa) = o((\log \kappa)^2)$.

Proof. By the Mertens estimate

$$A(x) := \sum_{p \leq x} \frac{\log p}{p} = \log x + O(1),$$

Abel summation gives

$$\sum_{p \leq \kappa} \frac{(\log p)^2}{p} = \log(\kappa) A(\kappa) - \int_2^\kappa \frac{A(t)}{t} dt = \frac{1}{2}(\log \kappa)^2 + O(\log \kappa).$$

Since $\sum_p f_p = D < \infty$ by Lemma 4.2, the identity $V_p(\frac{1}{2}) = 4(\log p)^2/p + f_p$ yields $H_{\text{local}}^{\text{ren}}(\frac{1}{2}, \kappa) = O(\log \kappa)$. $\square$

The leading constant 2 in $H_{\text{local}}(\frac{1}{2}, \kappa) \sim 2(\log \kappa)^2$ arises from $4 \cdot \frac{1}{2} = 2$.

Observation 5.2 (Sawtooth structure; proved). *For real $\kappa \geq 2$, the sum $\sum_{p \leq \kappa} V_p(\frac{1}{2})$ is a step function with jump $V_p(\frac{1}{2})$ at each prime p , while $2(\log \kappa)^2$ is continuous and strictly increasing. Hence $H_{\text{local}}^{\text{ren}}(\frac{1}{2}, \kappa)$ jumps upward at primes and decreases between consecutive primes.*

The *prime-power-tail energy* at cutoff κ is

$$H_{\text{tail}}(\kappa) := \sum_{p \leq \kappa} f_p = \sum_{p \leq \kappa} \frac{4(\log p)^2(2p-1)}{p(p-1)^2}. \quad (7)$$

By Lemma 4.2, $H_{\text{tail}}(\kappa) \nearrow D$ as $\kappa \rightarrow \infty$. The quantity H_{tail} is the partial sum of the diagonal energy: it is H_{tail} (not $H_{\text{local}}^{\text{ren}}$) that grows by f_p at each new prime and converges to D .

The same step-function mechanism as for $\pi(x)$ is present: $H_{\text{local}}^{\text{ren}}$ increases by $V_p(\frac{1}{2})$ at each prime $p \leq \kappa$ and decreases between consecutive primes as $2(\log \kappa)^2$ grows. The average drift of $H_{\text{local}}^{\text{ren}}$ is controlled by classical prime-sum estimates [6]:

$$\sum_{p \leq \kappa} \frac{\log p}{p} = \log \kappa + O(1) \quad (\kappa \rightarrow \infty).$$

The Mertens constant $\gamma_M \approx 0.2615$ governs the related estimate $\sum_{p \leq \kappa} 1/p = \log \log \kappa + \gamma_M + o(1)$, connecting the renormalised scale to classical prime-sum asymptotics.

Remark. *The Mertens constant*

$$\gamma_M = \gamma + \sum_p \left[\log\left(1 - \frac{1}{p}\right) + \frac{1}{p} \right] \approx 0.2615,$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. The constant γ_M governs the harmonic prime-sum asymptotic

$$\sum_{p \leq \kappa} \frac{1}{p} = \log \log \kappa + \gamma_M + o(1).$$

The renormalised curvature scale considered here is controlled instead by the weighted Mertens-type sums $\sum_{p \leq \kappa} (\log p)^2/p$ and $\sum_{p \leq \kappa} (\log p)/p$; thus γ_M is a related prime-sum constant, but it is not the leading constant governing $H_{\text{local}}^{\text{ren}}(\frac{1}{2}, \kappa)$. The same Euler–Mascheroni constant γ enters the first Li coefficient [5]

$$\lambda_1 = 1 + \frac{\gamma}{2} - \log 2 - \frac{\log \pi}{2} \approx 0.0231.$$

Remark. *The renormalised construction clarifies the scale of the prime-side contribution but does not by itself resolve the spectral inequality of Problem 7.2.*

6 Numerical comparison of zero and prime sides

We compare the renormalised zero sum with the prime-only renormalised local curvature on a finite grid in κ . For the numerical comparison, we use a simplified position-space proxy family

$$g_{\kappa,\varepsilon}^{\text{ren}}(x) := \sum_{p \leq \kappa} c_p^{\text{ren}} [\varphi_\varepsilon(x - \log p) - \varphi_\varepsilon(x + \log p)],$$

with weights $c_p^{\text{ren}} = f_p^{1/2}$ in place of the abstract weights $c_{p,\sigma}$ of the family $g_{\sigma,\varepsilon}^*$ constructed in §2. This proxy family differs from $g_{\sigma,\varepsilon}^*$ in its weights; Problem 7.2 concerns the original family.

In this section, the comparison is performed on the prime side only. Define the *prime-only renormalised curvature*

$$H_{\text{local}}^{\text{pr}}(\tfrac{1}{2}, \kappa) := \sum_{p \leq \kappa} V_p(\tfrac{1}{2}) - 2(\log \kappa)^2,$$

separating the archimedean contribution $\psi_1(\tfrac{1}{2}) \approx 2.150$, which enters the full renormalised curvature $H_{\text{local}}^{\text{ren}}(\tfrac{1}{2}, \kappa) = \psi_1(\tfrac{1}{2}) + \sum_{p \leq \kappa} V_p(\tfrac{1}{2}) - 2(\log \kappa)^2 = H_{\text{local}}^{\text{pr}}(\tfrac{1}{2}, \kappa) + \psi_1(\tfrac{1}{2})$ but is omitted from the grid comparison below. The renormalised zero sum is

$$Z_{\text{ren}}(\kappa) := Z_{\text{ord}}(g_{\kappa,\varepsilon}^{\text{ren}}; N),$$

computed at $\varepsilon = 0.05$, $N = 100$ zeros. The ordinate proxy $Z_{\text{ord}}(g_{\kappa,\varepsilon}^{\text{ren}}; N)$ is bounded at fixed ε and requires no subtraction of $2(\log \kappa)^2$; that renormalisation applies to the prime side $H_{\text{local}}^{\text{pr}}$ only. By the antisymmetric Gaussian construction and the Lagarias convention [3],

$$|\hat{g}_{\kappa,\varepsilon}^{\text{ren}}(\gamma_k)|^2 = 4e^{-\varepsilon^2 \gamma_k^2} \left(\sum_{p \leq \kappa} c_p^{\text{ren}} \sin(\gamma_k \log p) \right)^2, \quad (8)$$

which is non-negative and reproduces the phasor-damping factor $e^{-\varepsilon^2 \gamma_k^2}$ of the numerical computation.

The smoothing parameter $\varepsilon = 0.05$ is required for meaningful numerical experiments with the zero sum: for $\varepsilon = 0.5$ one has $e^{-\varepsilon^2 \gamma_1^2} \approx 10^{-22}$, effectively suppressing all zero contributions, so that no observable signal remains. With $c_p^{\text{ren}} = f_p^{1/2}$:

κ	Z_{ren}	$H_{\text{local}}^{\text{pr}}$	$Z_{\text{ren}} - H_{\text{local}}^{\text{pr}}$	$Z_{\text{ren}} \geq H_{\text{local}}^{\text{pr}}$
10	4.888	3.044	+1.844	✓
20	3.782	4.770	−0.988	×
29	5.520	3.588	+1.931	✓
50	5.693	2.881	+2.812	✓
100	5.311	1.562	+3.748	✓
200	5.102	2.355	+2.747	✓
500	5.156	1.101	+4.056	✓
1000	4.289	0.426	+3.863	✓

Observation 6.1 (Finite-grid stability of the comparison; numerical). *On the eight-point grid $\kappa \in \{10, 20, 29, 50, 100, 200, 500, 1000\}$, the inequality $Z_{\text{ren}} \geq H_{\text{local}}^{\text{pr}}$ holds at every tested point except $\kappa = 20$, where $Z_{\text{ren}} - H_{\text{local}}^{\text{pr}} = -0.988$. The ratio $Z_{\text{ren}}/H_{\text{local}}^{\text{pr}}$ equals about 1.5 at $\kappa = 29$ and about 10.1 at $\kappa = 1000$. No claim is made about κ outside the eight tested points.*

The single grid violation $Z_{\text{ren}} < H_{\text{local}}^{\text{pr}}$ at $\kappa = 20$ illustrates the sawtooth behaviour of $H_{\text{local}}^{\text{pr}}$: directly after a prime jump $H_{\text{local}}^{\text{pr}}$ rises by $V_p(\frac{1}{2})$, then decreases continuously until the next prime. The pattern observed on the grid for $\kappa \geq 29$ is consistent with the spectral inequality of Problem 7.2, but does not prove it: Problem 7.2 concerns the unconditional Lagarias–Weil zero-side contribution $\mathcal{Z}_W(g_{1/2,\varepsilon}^*)$, whereas the present computation uses the finite positive-ordinate proxy Z_{ord} . No positivity or inequality for $\mathcal{Z}_W$ is inferred from the finite-grid data.

All numerical claims of this section are reproducible from the verification scripts in the GitHub repository [2].

7 Open Problems

The present work identifies a formal diagonal localisation pattern (Observation 3.2) and reduces the question of prime-side derivation of Weil positivity for the family $g_{\sigma,\varepsilon}^*$ to three well-posed subproblems. We name each and indicate the tools we expect to be relevant.

Open Problem 7.1 (Weil-normalised embedding). *Determine whether the normalised diagonal localisation of Observation 3.2 can be implemented inside the Lagarias–Weil functional [3] for the position-space family $g_{\sigma,\varepsilon}^*$, including:*

- (a) *the correct Gaussian rescaling so that the convolution value at the origin matches the prime-side Euler factor $V_p(\sigma)$;*
- (b) *the archimedean Weil-factor contribution $\mathcal{A}_\infty(g_{\sigma,\varepsilon}^* * \check{g}_{\sigma,\varepsilon}^*)$, including the identification and, if necessary, subtraction or rescaling of the singular self-correlation term $[\varphi_\varepsilon * \check{\varphi}_\varepsilon](0) \sim (2\sqrt{\pi}\varepsilon)^{-1}$ as $\varepsilon \rightarrow 0$;*
- (c) *the zero-side contribution $\mathcal{Z}_W(g_{\sigma,\varepsilon}^*)$ in its unconditional form, without assuming critical-line localisation.*

A positive answer would establish the formal prime-side identity

$$W(g_{\sigma,\varepsilon}^* * \check{g}_{\sigma,\varepsilon}^*) = \mathcal{Z}_W(g_{\sigma,\varepsilon}^*) - H_{\text{local}}(\sigma, \kappa) + R(\sigma, \varepsilon, \kappa)$$

as a proved statement, with $R \rightarrow 0$ as $\varepsilon \rightarrow 0$ at fixed (σ, κ) .

For the global (adèle) setting, Connes and Consani [12] establish Weil positivity at the archimedean place via a trace formula on Sonin's space, using prolate spheroidal wave functions and Toeplitz matrix theory; the UV prolate spectrum is further developed in [13]. Our finite-cutoff Gaussian family $g_{\sigma,\varepsilon}^* \in S_{\text{ad}}$ operates in the Lagarias local framework at fixed κ ; the archimedean limit as $\varepsilon \rightarrow 0$ in this setting remains the content of (b). Thus [12] provides an archimedean positivity precedent, but not a direct evaluation of $\mathcal{A}_\infty(g_{\sigma,\varepsilon}^* * \check{g}_{\sigma,\varepsilon}^*)$ for the finite-cutoff Gaussian family used here; in particular, it does not control the $\varepsilon \rightarrow 0$ singular self-correlation term in the Lagarias normalisation.

Open Problem 7.2 (Spectral inequality on the critical line). *Prove that for every cutoff κ there exists $\varepsilon_0(\kappa) > 0$ such that for all $0 < \varepsilon < \varepsilon_0(\kappa)$,*

$$\mathcal{Z}_W(g_{1/2,\varepsilon}^*) \geq H_{\text{local}}(\tfrac{1}{2}, \kappa), \quad (9)$$

where $\mathcal{Z}_W$ denotes the unconditional zero-side contribution in the Lagarias–Weil functional (Problem 7.1(c)). This inequality would be a necessary component of a proof of Weil positivity through this specific family, after the embedding problem of Problem 7.1 has been resolved. By itself it is not an RH criterion and is not asserted to be necessary for RH in general. By Lagarias [3], the equivalence $\text{RH} \iff W(f * \check{f}) \geq 0$ holds for all $f \in S_{\text{ad}}$. We note that Bombieri [15] analyses Weil's quadratic functional on support-restricted L^2 -spaces and finite-dimensional truncations. This illustrates that positivity is a delicate statement about the chosen test-function class and limiting process: on the full relevant admissible class, Weil positivity is RH-equivalent, not a formal consequence of the local sign pattern or of the diagonal localisation in Observation 3.2. The expected tools are: explicit lower bounds on the zero-side quadratic form by the unconditional zero-counting density; finite- κ cross-term estimates for the original family $g_{1/2,\varepsilon}^*$. Problem 7.3 formulates a related but distinct proxy-family question in the joint regime $\kappa \rightarrow \infty$, $\varepsilon = \varepsilon(\kappa) \rightarrow 0$. On the tested grid the inequality $Z_{\text{ren}} \geq H_{\text{local}}^{\text{pr}}$ holds at every tested point except $\kappa = 20$ (Observation 6.1), consistent with (9) but not establishing it analytically.

Open Problem 7.3 (Off-diagonal control of the zero sum). *To avoid conflict with the finite numerical proxy $Z_{\text{ord}}(g; N)$ used in § 6, we write $Z_+^\infty(g)$ for the infinite positive-ordinate quadratic form obtained by summing over all $\gamma_k > 0$. Thus $Z_{\text{ord}}(g; N)$ is finite and numerical, whereas $Z_+^\infty(g)$ is the asymptotic object in this problem.*

Prove unconditionally that the combined non-diagonal ratio contribution and product-term remainder in the positive-ordinate zero sum are asymptotically negligible compared with the diagonal R_1 contribution. Set

$$I_+(\varepsilon) := \sum_{\gamma_k > 0} e^{-\varepsilon^2 \gamma_k^2}.$$

By the Riemann–von Mangoldt zero-counting formula and partial summation,

$$I_+(\varepsilon) = \frac{1}{4\sqrt{\pi}\varepsilon} \log \frac{1}{\varepsilon} + O\left(\frac{1}{\varepsilon}\right) \quad (\varepsilon \rightarrow 0^+).$$

Thus, for $\varepsilon(\kappa) \sim (4\sqrt{\pi} \log \kappa)^{-1}$, one has $I_+(\varepsilon(\kappa)) \sim \log \kappa \log \log \kappa$. For

$$R_a(\varepsilon) := \sum_{\gamma_k > 0} e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log a),$$

one has $R_1(\varepsilon) = I_+(\varepsilon)$. The identity

$$Z_+^\infty(g_{\kappa,\varepsilon}^{\text{ren}}) = 2 \sum_{p,q \leq \kappa} c_p^{\text{ren}} c_q^{\text{ren}} R_{p/q}(\varepsilon) - 2 \sum_{p,q \leq \kappa} c_p^{\text{ren}} c_q^{\text{ren}} R_{pq}(\varepsilon)$$

follows from $4 \sin(\gamma \log p) \sin(\gamma \log q) = 2 \cos(\gamma \log(p/q)) - 2 \cos(\gamma \log(pq))$. The principal diagonal term is

$$2 \sum_{p \leq \kappa} (c_p^{\text{ren}})^2 I_+(\varepsilon) = 2 H_{\text{tail}}(\kappa) I_+(\varepsilon).$$

Prove unconditionally that, in the expansion of $Z_+^\infty(g_{\kappa,\varepsilon}^{\text{ren}})$ above, for the scale

$$\varepsilon(\kappa) \sim (4\sqrt{\pi} \log \kappa)^{-1},$$

the combined non-diagonal and product-term remainder satisfies

$$2 \sum_{\substack{p,q \leq \kappa \\ p \neq q}} c_p^{\text{ren}} c_q^{\text{ren}} R_{p/q}(\varepsilon) - 2 \sum_{p,q \leq \kappa} c_p^{\text{ren}} c_q^{\text{ren}} R_{pq}(\varepsilon) = o(H_{\text{tail}}(\kappa) I_+(\varepsilon)).$$

Then, since $H_{\text{tail}}(\kappa) \rightarrow D$ (Lemma 4.2), one obtains

$$Z_+^\infty(g_{\kappa,\varepsilon(\kappa)}^{\text{ren}}) \sim 2D I_+(\varepsilon(\kappa)).$$

For example, if $\varepsilon(\kappa) \sim (4\sqrt{\pi} \log \kappa)^{-1}$, then $I_+(\varepsilon(\kappa)) \sim \log \kappa \log \log \kappa$ and hence

$$Z_+^\infty(g_{\kappa,\varepsilon(\kappa)}^{\text{ren}}) \sim 2D \log \kappa \log \log \kappa,$$

motivated by Selberg-type orthogonality and mean-square considerations [7]. The expected tools are: analytic continuation of $R_a(\varepsilon)$ for $a \neq 1$ via classical explicit-formula techniques; cancellation by oscillation, with the R_{pq} product terms providing additional cancellation. The Beurling–Selberg extremal function framework used by Carneiro–Chandee–Littmann–Milinovich [19] and the semidefinite programming methods of Chirre–Gonçalves–de Laat [20] provide methodological precedents for bounding zero-statistical sums with Gaussian test functions. They are cited here as analytic tools to be adapted, not as hypotheses or as unconditional inputs to Problem 7.3. This problem is formulated entirely within classical analytic number theory.

Remark. Problems 7.1, 7.2, and 7.3 address different regimes. Problem 7.1 concerns the structural embedding of the diagonal localisation into the Lagarias–Weil functional. Problem 7.2

is a fixed-cutoff, small- ε inequality: for each finite κ , it asks whether there exists $\varepsilon_0(\kappa) > 0$ such that the inequality holds for all $0 < \varepsilon < \varepsilon_0(\kappa)$. Problem 7.3 concerns the joint limit $\kappa \rightarrow \infty$, $\varepsilon = \varepsilon(\kappa) \rightarrow 0$, and addresses the asymptotic structure of the ordinate sum in that regime. The diagonal scale $2D \log \kappa \log \log \kappa$ in Problem 7.3 is asymptotically smaller than $H_{\text{local}}(\frac{1}{2}, \kappa) \sim 2(\log \kappa)^2$; the two problems are therefore complementary rather than directly linked.

Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward.

No RH as input. At no point is the Riemann Hypothesis assumed. The non-trivial zeros ρ enter the construction only through the classical zero-counting density $N(T) \sim T \log T / (2\pi)$, which is unconditional, and through finite Gaussian-smoothed sums for which the symmetric pairing $\rho \leftrightarrow \bar{\rho}$ is used without further hypothesis. The reference point $\sigma = \frac{1}{2}$ is chosen as the object of study, not assumed to be the unique zero location.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. The construction of $g_{\sigma, \varepsilon}^*$ uses only Schwartz-class smoothing and antisymmetrisation; it makes no reference to random-matrix statistics, pair-correlation conjectures, or any postulated spectral realisation of the zeros. The diagonal scale in Problem 7.3 is obtained by combining $D = \sum_p f_p$ with the unconditional Riemann–von Mangoldt zero-counting density; no GUE, Montgomery pair-correlation, or Hilbert–Pólya input is used.

The Weil explicit formula is classical. The diagonal localisation (Observation 3.2) is an algebraic identity in the weights $c_{p, \sigma}$, independent of the Weil functional. The embedding into the Lagarias–Weil framework [3] is formulated as an open problem (Problem 7.1); no unverified identity is used.

Numerical results are clearly marked. The diagonal energy value $D \approx 9.470$ and the comparison-grid statements of §6 rely on direct computation; each statement is labelled [numerical] in its heading.

Open problems are named, not used as hypotheses. Problems 7.2 and 7.3 are stated openly in §7. No claim of this paper is conditional on a positive answer to either problem.

Acknowledgments

The author thanks Jean-François Burnol for confirming that the curvature decomposition of [1] does not appear in the literature known to him, and the readers who provided feedback on earlier versions of this paper.

Call for Feedback

This paper is part of an open research initiative on prime–zero coupling and explicit-formula operators. All papers in the series are freely available on Zenodo under CC BY 4.0, and the accompanying verification code is hosted on GitHub.

The author welcomes critical feedback, corrections, and collaboration:

Zenodo:

https://zenodo.org/search?q=metadata.creators.person_or_org.name%3A%22Tehrani%20Ulrich%22

GitHub:

<https://github.com/utehrani/analysislab-nt>

Topics of particular interest include: analytical approaches to the spectral inequality of Problem 7.2; unconditional control of the off-diagonal cross-terms of Problem 7.3; and the relation between the diagonal constant D in (6) and the classical zero-counting density entering $I_+(\varepsilon)$.

We particularly welcome expert comment on the following question: for the Gaussian-smoothed proxy family $g_{\kappa,\varepsilon}^{\text{ren}}$ used in §6, is the asymptotic comparison

$$Z_+^\infty(g_{\kappa,\varepsilon(\kappa)}^{\text{ren}}) \sim 2D \log \kappa \log \log \kappa, \quad \varepsilon(\kappa) \sim (4\sqrt{\pi} \log \kappa)^{-1},$$

with D as in (6), accessible by classical sieve or explicit-formula techniques without recourse to random-matrix statistics?

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